

Andrew W. Shenstone

Shenstone, Andrew - Application for Staff Software Engineer (Payer) — Payers & Outcomes @ Headway

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SUMMARY

Staff-shaped engineer with twelve years shipping end-to-end product: **TypeScript**, **Next.js**, **PostgreSQL**, and production LLM features users actually rely on. I like hard problems where the interface, the API, and the data model all have to stay honest, especially when the stakes are trust and safety, not just click-through.

Payers & Outcomes is the kind of mandate I want: intelligent systems over real clinical and claims volume, feedback loops that help providers instead of punishing them, and evaluation discipline so models don't become liability dressed up as automation. My proof is Plantfolio (founder-lead, full stack, AI in the loop) plus prior chapters on risk-adjacent tooling, multi-tenant data rules, and mentoring without needing a manager title.

I have not worked inside U.S. insurance claims, billing, or HIPAA-scoped healthcare stacks—that is squarely a domain ramp I'd treat with respect. My transfer is deep full-stack execution, regulated-adjacent fintech tooling (Tilia), and a finance degree that helps me read payer and utilization problems without glazing over.

SKILLS

LLM-powered product systems: Shipped user-facing LLM features in production: vision and language models, structured extraction, and pragmatic model choice (Plantfolio), AI SDK and OpenAI APIs; bias toward evaluation, iteration, and observability so features stay reliable—not one-off demos, Comfortable with the "what to automate vs keep human" tradeoff in sensitive workflows; eager to deepen formal eval loops for regulated healthcare AI

Full-stack & data-heavy backends: TypeScript end to end on Next.js and Node; PostgreSQL with Prisma and thoughtful schema design in production, Python is not my heaviest production lane today; I ramp quickly across languages and have used Django in past work—FastAPI and SQLAlchemy are a focused study path, not a blank slate, GraphQL schema-through-client experience (Bloom, Tilia); API design, debugging across layers, and ownership from problem to deployed fix

Frontend for complex, trusted workflows: React and Next.js with server components, disciplined data fetching, and performance habits that matter in long-session products, Remix is not in my production history yet; same mental model as modern React loaders and server boundaries—I learn your house patterns quickly, Dense admin and operator surfaces (Bloom reporting/admin; Plantfolio); accessible, maintainable UI with tests where they buy the most safety

Leadership craft (no direct reports): Informal technical lead through pairing, design discussion, and code review that teaches patterns—not just blocks merges, Founder chapter on Plantfolio: ambiguous scopes, tight iteration, clear async updates across stakeholders, Established standards, component libraries, and CI-backed quality so teams move fast without sharp edges

Cloud & Tools: AWS (S3, EC2), Vercel, Algolia, Inngest, Clerk Auth, Stripe, Git, CircleCI, CI/CD, pnpm, Turbo, Cursor IDE (Top 1% user)

EDUCATION

B.S. Finance, Santa Clara University — 2010

EXPERIENCE

Founder & Lead Engineer

Plantfolio | November 2023 - Present

Product Leadership & Architecture

- Architected and built a comprehensive plant management platform from the ground up, serving as both technical lead and product owner

- Owned the full feature cycle: joined customer calls to identify friction points, architected solutions, implemented, tested, and verified success in production
- Designed and implemented a sophisticated monorepo architecture with 12+ packages, enabling rapid development and maintainable codebases
- Led product strategy and user experience decisions, making product micro-decisions and navigating ambiguity to deliver value iteratively

Technical Achievements

- **AI Integration:** Built advanced AI-powered features including plant identification from photos, contextual chatbot (Sage), and smart text extraction from plant tags using OpenAI's vision and language models
- **Modern Frontend Architecture:** Implemented Next.js with React, React Server Components, and TypeScript, achieving exceptional performance and developer experience
- **Complex Data Modeling:** Designed and implemented sophisticated PostgreSQL schemas with Prisma, handling complex relationships between users, plants, folios, and media with 15+ interconnected models
- **Supabase to Neon:** Shipped on Supabase (PostgreSQL with Row-Level Security policies for multi-tenant isolation) before migrating the data layer to Neon; strong hands-on experience with RLS and Postgres-centric access control
- **Content Management:** Created sophisticated content management system with rich text editing, plant tagging, and flexible data structures

Scale & Performance

- Managed a database of 40,000+ plant species with advanced search capabilities using Algolia and vector embeddings
- Implemented comprehensive media management system with AWS S3, supporting image optimization, galleries, and metadata extraction
- Built robust authentication and authorization systems with Clerk, supporting multiple user types and subscription tiers
- Optimized application performance through React Server Components, efficient data fetching, and strategic caching

Client Case Study - AI Signage

- Delivered a production rollout for an independent garden center to automate plant signage generation; one independent retail customer in ongoing production who has been an enthusiastic reference
- Enabled bulk plant import (plain text) to printable signs in seconds; reduced a weekend-long manual process to minutes after initial setup
- Demonstrated ability to translate customer requirements into scalable technical solutions and shorten feedback loops

Full-Stack Ownership

- Developed complete payment integration with Stripe, subscription management, and user onboarding flows
- Built comprehensive email system with React Email, supporting transactional emails, notifications, and marketing campaigns
- Implemented advanced search and filtering capabilities with Algolia, supporting complex plant queries and user preferences

Code Quality & Leadership

- Established coding standards and architectural patterns that enabled rapid, consistent development
- Implemented comprehensive testing strategy with Vitest and Cypress, maintaining high code quality
- Built reusable component library and design system, ensuring consistent UI/UX across the platform

Senior Software Engineer

Tilia | January 2023 - August 2023

- Partnered with cross-functional teams in product and design to revamp critical user-facing features

- Built and shipped work in support of KYC, identity verification, and risk-adjacent tooling as part of Tilia's payments platform (tools team context)
 - Introduced and evolved a GraphQL API on the platform while consuming it from our product—hands-on with both server-side schema/resolvers and client integration
 - Spearheaded and guided a team through the successful adoption of Chakra UI, reducing bespoke components and establishing cohesive design patterns
 - Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
 - Maintained exceptional unit test coverage of over 90% for the Tools team application, alongside comprehensive end-to-end testing
 - Led the swift creation and deployment of a homepage reporting dashboard within the initial two months of joining
 - Role ended in a company-wide layoff affecting approximately one-third of employees
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Senior Software Engineer

Bloom.wine | September 2020 - October 2022

- Architected a greenfield Next.js application for the primary admin interface of the Bloom product; much of that stack remains in production across many winery clients today
- Engineered Ruby on Rails GraphQL API endpoints shared by multiple applications across mobile and web devices
- CI/CD with CircleCI for build, test, and deployment workflows alongside the Next.js and Rails stack
- Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
- Collaborated with designers using Figma, establishing component library standards usable by both engineering and design
- Integrated DatoCMS and extended application functionality through third-party API integrations (Shopify, Mailchimp, Stripe, Shipstation, ActiveCampaign); all Bloom

winery clients were on Shopify Plus

Software Engineer II

Tune Inc - Marketplace Automation | August 2019 - May 2020

- Built and operated ETL-style data pipelines and integrations supporting marketplace automation workflows
 - Automated a client workflow using a Node.js/Express API and React.js, reducing the time to complete the task from 10-15 minutes to 10-30 seconds
 - Modernized the primary legacy React.js application by moving from a custom Webpack process to react-scripts, resulting in a 50% reduction in production build size and a 60% reduction in NPM dependencies
 - Reduced development machine CPU usage from 75% to <5% by simplifying the build pipeline and dependency graph
 - Role ended in a company-wide layoff
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Software Developer

Ply Interactive Inc | June 2014 - July 2019

React Native

- Shipped multiple production React Native applications for agency clients—cross-platform mobile apps delivered end to end
- Washington State Fair: React Native app in sustained multi-year production; very high traffic during fair season, from tens of thousands of users into the hundreds of thousands cumulatively over seasons

Web & platforms

- Developed numerous web applications including Georgetownbeer.com, WashingtonWine.org, TwoWolvesWine.com, Novaerus.com, WellAir, and Plasma-air.com

- Built a standalone Ruby gem Content Management System, enabling simplified content management within Ruby on Rails projects
- Transitioned the agency to modern technologies like GraphQL, Gatsby.js, and React.js, improving delivery times for new projects
- Bloom.wine Product Development: Built extensive reporting features using complex PostgreSQL queries and interactive filtering, saving clients thousands of hours on previously manual tasks
- Engineered a flexible Wine Club membership structure for wineries, allowing for diverse client configuration and broadening the range of potential sales targets
- Communicated and enforced the standards of Shopify's Polaris React.js component library with designers, resulting in efficiencies gained from leveraging pre-built components

COMPANY REFERENCE

Plantfolio (2023-Present) - AI-powered plant management platform with 40,000+ plant database serving both consumer garden management and independent garden center services - plantfolio.garden/

Tilia (2023) - All-in-one payment platform for digital economies - tilia.io/

Bloom.wine (2020-2022) - Commerce services for the alcohol industry; all clients on Shopify Plus - bloom.wine/

Tune Inc - Marketplace Automation (2019-2020) - SaaS platform for managing marketing partnerships across mobile and web - tune.com/

Ply Interactive Inc (2014-2019) - Boutique agency: web and production React Native apps (e.g. Washington State Fair at major scale); developed Bloom.wine product internally (2017-2019)